BIOBUZZ Autonomous Routes

One page per autonomous OpMode, drawn from the JSON route files the robot runs and the Pedro paths it follows. Points and times are estimates; the checks catch rule and field problems before the robot does.

OPMODE	ROUTE FILE	EST. POINTS	EST. TIME	CHECKS
Red: Leave + Park	leave_and_park.json	**3 pts**	4.2 s	**1 problem (see page)**
Blue: Leave + Park	leave_and_park.json	**3 pts**	4.2 s	**3 problems (see page)**

HOW TO READ A PAGE

- Path the robot follows, in step order; the chevron points the way it drives.
- Robot at START (18 x 18 in max, R102); the thick edge is its front.
- Robot after each path.
- Robot at the end of AUTO, where LEAVE and PARK are judged.
- 2 — Routine step number, matching the list on the page.
- Waypoint the robot passes through; hollow means a Bezier control point it doesn't reach.

WHAT THE NUMBERS ASSUME

Field: the Pedro Pathing visualizer's BIOBUZZ field, 141.5 in square. Origin at the red GARDEN corner, x toward the blue wall, y away from the audience, heading 0° = facing +x.

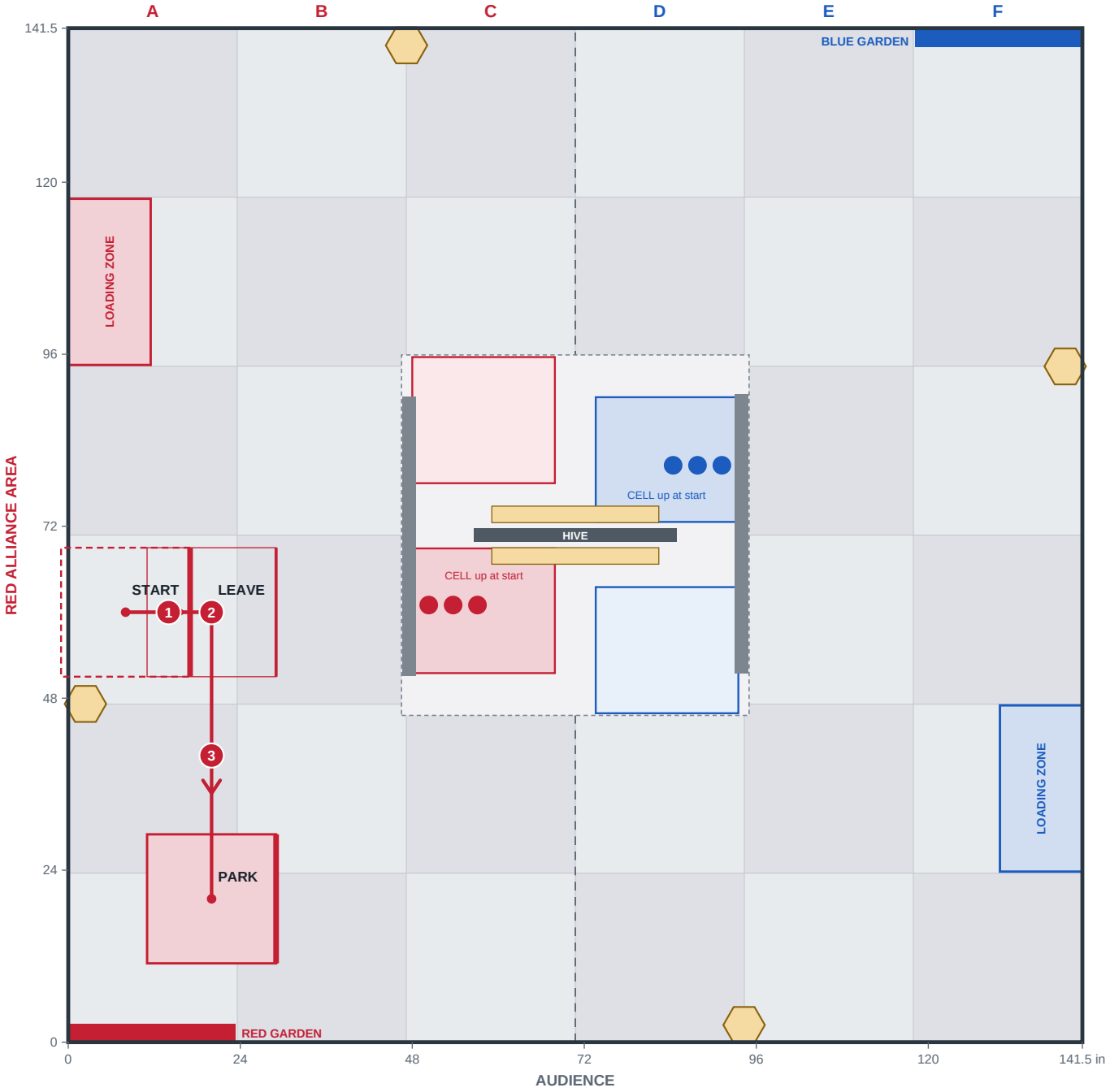
Robot: 18 x 18 in. Time per path: accelerate at 37 in/s² to 50 in/s, decelerate, then 0.25 s to settle. Action and waitUntil times aren't known (shown as ?).

Points: LEAVE 3 if the robot ends clear of every wall; PARK 5 if it ends partly in its own LOADING ZONE; HIVE tips aren't estimated until routes have launch actions.

Routes come from TeamCode/src/main/assets/autos. A copy pushed to the Control Hub (/sdcard/FIRST/autos) overrides it on the robot but isn't drawn here.

Red: Leave + Park

leave_and_park.json · route “Leave + Park” · LeaveAndParkAuto.Red · then TeleOp “Red: TeleOp” · group comp



ESTIMATED POINTS

3 points

LEAVE +3
ends 11 in from the nearest wall

PARK 0
ends outside the red LOADING ZONE (x 0.0–11.5, y 94.5–117.7)

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

4.2 s of 30 s

CHECKS

- START puts the robot 1.0 in into the wall (robot 18 x 18 in). Move START so the robot just touches the wall (G304).

ROUTE

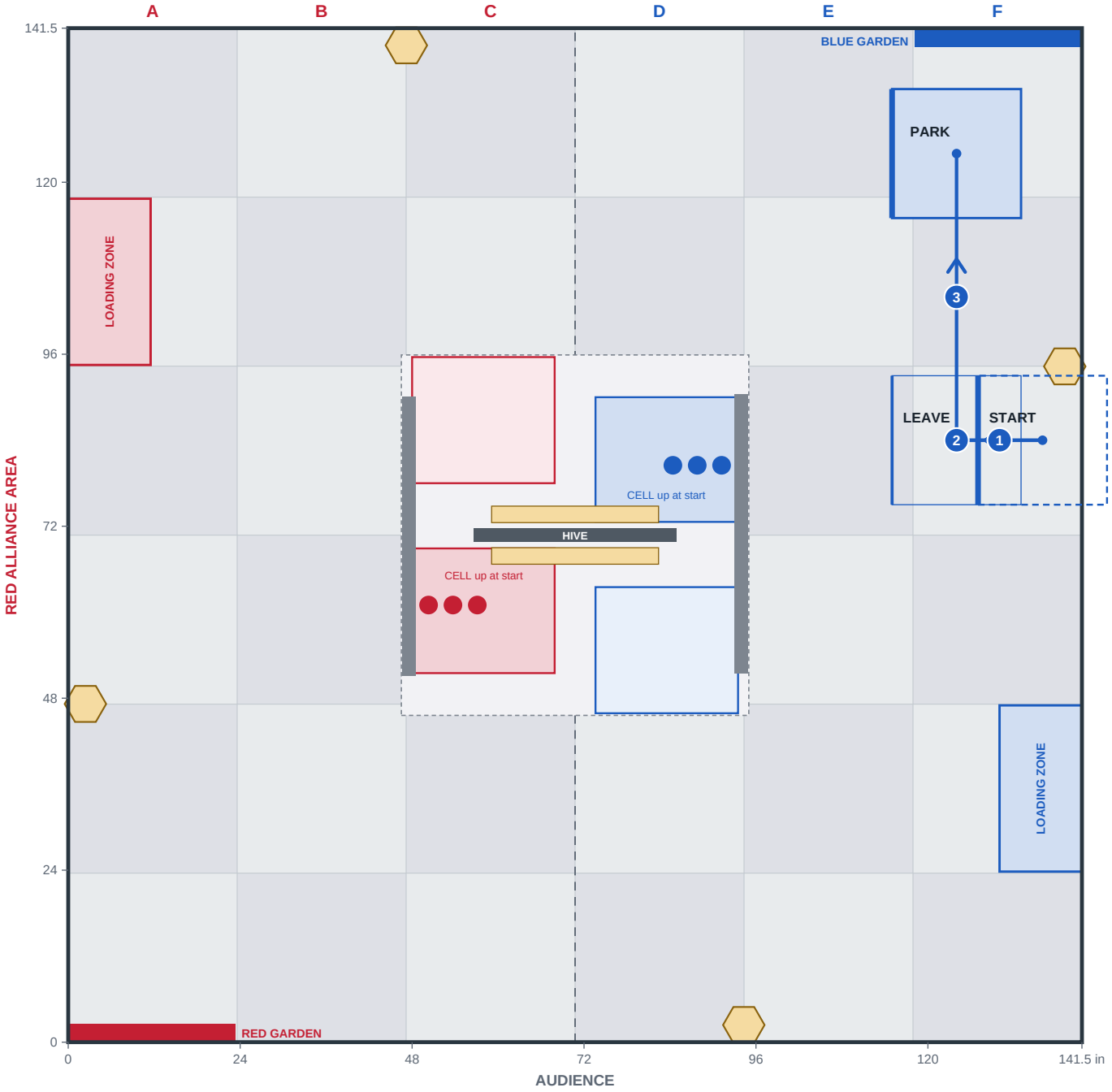
1	leave wall: line START -> LEAVE, 12 in	~1.4 s
2	wait 0.5 s	~0.5 s
3	park: line LEAVE -> PARK, 40 in	~2.3 s

WAYPOINTS, RED FRAME (x, y, heading)

START	8.0, 60.0, 0°
LEAVE	20.0, 60.0, 0°
PARK	20.0, 20.0, 0°

Blue: Leave + Park

leave_and_park.json · route “Leave + Park” · LeaveAndParkAuto.Blue · then TeleOp “Blue: TeleOp” · group comp



ESTIMATED POINTS

3 points

LEAVE +3
ends 9 in from the nearest wall

PARK 0
ends outside the blue LOADING ZONE (x 130.0–141.5, y 23.8–47.0)

HIVE TIPS 0
none: the route has no launch actions

ESTIMATED TIME

4.2 s of 30 s

CHECKS

- START puts the robot 3.5 in into the wall (robot 18 x 18 in). Move START so the robot just touches the wall (G304).
- START touches a FLOWER (G304).
- Alliance.BLUE rotates about (72, 72), but the field centre is (70.75, 70.75): every blue point is 2.5 in off in x and y.

ROUTINE

1	leave wall: line START -> LEAVE, 12 in	~1.4 s
2	wait 0.5 s	~0.5 s
3	park: line LEAVE -> PARK, 40 in	~2.3 s

WAYPOINTS, BLUE FRAME (x, y, heading)

START	136.0, 84.0, 180°
LEAVE	124.0, 84.0, 180°
PARK	124.0, 124.0, 180°